

BOULMER, United Kingdom

WMO: 032400

Lat: 55.4208N

Lon: 1.6014W

Elev: 21

StdP: 101.07

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.8	-0.6	-5.8	2.3	0.4	-4.7	2.6	1.1	18.1	8.9	15.8	7.8	4.0	260	0.687

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.4	21.0	16.3	19.7	15.6	18.6	15.0	17.4	19.8	16.6	18.7	15.8	17.8	5.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	11.7	18.5	15.7	11.1	17.7	14.9	10.6	16.9	49.0	19.9	46.5	18.7	44.4	17.9	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.7	12.1	10.6	DB	-4.2	24.2	2.1	0.9	-5.7	24.8	-6.9	25.4	-8.1	25.9	-9.6	26.6	
			WB	-5.0	19.4	1.9	1.0	-6.4	20.1	-7.5	20.7	-8.6	21.3	-10.0	22.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	4.7	4.9	5.9	7.5	9.9	12.6	14.7	14.8	13.0	10.4	7.2	5.0
	DBStd	4.40	2.48	2.60	2.58	2.48	2.39	2.16	1.98	1.86	2.01	2.35	2.60	3.00
	HDD10.0	823	165	144	129	82	31	2	0	0	2	23	89	156
	HDD18.3	3319	423	376	385	325	261	172	113	112	160	245	334	413
	CDD10.0	548	1	1	2	7	28	81	147	148	92	36	5	1
	CDD18.3	2	0	0	0	0	0	1	1	1	0	0	0	0
	CDH23.3	4	0	0	0	0	0	1	2	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.0	6.2	6.1	5.7	4.9	4.4	4.2	3.8	4.0	4.5	5.2	5.6	5.9
Precipitation	PrecAvg	724	60	48	42	54	49	62	65	67	56	72	82	67
	PrecMax	1030	159	117	98	156	100	163	161	152	134	169	191	139
	PrecMin	540	17	6	9	8	11	14	11	11	16	18	26	25
	PrecStd	125	31	27	25	42	24	38	35	31	32	31	36	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	13.1	14.7	16.9	19.0	21.8	23.1	22.7	21.7	18.0	15.1	13.7
		MCWB	10.4	10.6	10.9	12.6	13.9	16.3	17.5	17.3	16.7	14.8	13.0	11.7
	2%	DB	11.1	11.6	12.8	14.6	16.9	19.3	21.0	20.6	19.2	16.2	13.3	11.8
		MCWB	9.1	9.5	9.6	10.8	12.7	14.7	16.4	16.3	15.4	13.5	11.3	10.0
	5%	DB	9.9	10.4	11.4	13.1	15.3	17.7	19.6	19.4	17.7	15.0	12.2	10.8
		MCWB	8.0	8.2	8.6	9.6	11.7	13.7	15.6	15.6	14.3	12.7	10.3	9.1
	10%	DB	8.6	9.2	10.2	11.6	13.9	16.5	18.5	18.2	16.4	13.9	11.2	9.4
		MCWB	6.9	7.1	7.7	8.8	10.9	13.1	15.1	15.1	13.6	11.9	9.7	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	11.0	11.4	13.3	15.0	17.2	18.5	18.6	17.6	15.4	13.3	11.8
		MCDB	12.1	12.6	14.0	16.1	17.8	20.5	21.6	20.9	20.6	17.1	14.7	13.5
	2%	WB	9.3	9.6	10.1	11.4	13.6	15.5	17.4	17.4	16.0	14.1	11.8	10.4
		MCDB	10.9	11.4	12.2	14.1	16.0	18.2	19.9	19.5	18.4	15.6	12.9	11.7
	5%	WB	8.2	8.5	9.0	10.2	12.3	14.5	16.5	16.6	15.0	13.2	10.8	9.3
		MCDB	9.5	10.2	11.1	12.3	14.5	16.8	18.8	18.4	16.7	14.5	11.7	10.6
	10%	WB	7.2	7.4	7.9	9.1	11.4	13.7	15.7	15.9	14.2	12.4	10.0	8.1
		MCDB	8.3	8.9	9.9	11.1	13.4	15.8	17.8	17.5	15.8	13.6	10.9	9.3
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.1	5.8	5.9	6.2	6.1	6.5	6.4	6.3	5.5	5.0	4.6
		MCDBR	6.1	6.3	7.6	8.3	8.4	8.5	8.4	8.0	8.1	6.7	6.1	6.0
		MCWBR	5.5	5.1	5.4	5.4	5.1	5.1	4.9	4.8	5.2	4.9	5.2	5.5
	5% WB	MCDBR	5.9	6.2	7.3	7.8	7.6	7.6	7.5	6.8	7.2	6.2	5.5	5.9
		MCWBR	5.4	5.3	5.7	5.5	5.1	5.0	5.0	4.9	5.3	5.1	5.0	5.5
Clear-Sky Solar Irradiance	taub		0.317	0.334	0.365	0.389	0.385	0.387	0.389	0.379	0.360	0.344	0.318	0.300
	taud		2.307	2.382	2.320	2.284	2.348	2.368	2.370	2.409	2.463	2.457	2.365	2.271
	Ebn at Noon		599	741	797	831	858	858	849	834	802	722	597	523
	Edh at Noon		55	73	97	115	114	114	111	101	84	67	51	45
All-Sky Solar Radiation	RadAvg		0.54	1.18	2.36	3.90	5.08	5.04	4.81	4.01	2.83	1.52	0.70	0.40
	RadStd		0.07	0.12	0.28	0.35	0.27	0.39	0.36	0.27	0.23	0.12	0.06	0.04

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.26	N/A	N/A	N/A	N/A	N/A	N/A	-91	+47	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page